Business Workflow specifications

LLM version: Gemini Pro

PROMPT:

Role : You are an event stream generator.

To do: Using only the provided system specification, independently generate 40 event records.

Requirements:

1. Follow all defined event types, schemas, workflow rules, and temporal constraints.
2. Generate all event attribute values independently.
3. Sample event outcomes according to the specified probability distributions and dependencies.
4. Maintain consistency across related events and database identifiers.
5. Do not use existing event records or templates.
6. Output only the generated event records in the specified format.

Specifications:

Research Overview:

The specification determines what a valid event stream must satisfy. The LLM independently determines the actual event occurrences, identifiers, timestamps, selected entities and outcomes. The model is not supposed to know CEPAL. All requirements must be understandable without access to its syntax, implementation, or generated datasets.

System modeled:

The Multi-Delivery App is an application where students can order food, books, and other supplies from one or more sellers in a single order. A student selects products, places an order, pays and receives the order through a driver. Each seller prepares its own portion of the order. The driver must collect all required portions before the order can be marked delivered. A failed payment results in cancellation of the order.

Units used:

Event: A typed, time-stamped record of an action or state transition

Case(order): All events sharing one order_id, the order_id is the process-instance identifier

Sellers subcase: Events for one (order_id, seller_id) pair, these form a seller-specific preparation and pickup sequence.

Event stream: An ordered collection of events for multiple cases, potentially interleaved in event time.

Generated dataset: The event stream plus any required entity/ relationship data used to validate identifiers and payloads.

Every degenerated event must be identifiable, associated with the correct process instance, and placed at a valid time. Events may carry additional attributes depending on their type.

Simulation parameters:

Time unit: Minutes

Time granularity: 1 minute

Nominal simulation period: 720 minutes

Reference clock: 08:00-20:00

Event times: Integer minutes offsets

Order-start interval: From minute 0 up to, but not including, minute 720

Concurrent processes: Allowed

Generated event values: Not supplied in advance

What this document is not:

This document does not contain example traces for copying. It does not instruct an LLM to imitate rows from the classical generator. It does not assert that ambiguous CEPAL clauses have single obvious interpretation. It does not specify a programming method for producing the output.

Database and resource specifications:

The system consist of six logical database tables

Table	Required attributes	Purpose
Users	User_id, role	Students and drivers
Sellers	Seller_id, category	Stores and restaurants
Items	item_id , seller_id, is_available	Products associated with sellers
Orders	order _id, student_id, driver_id, delivery_location, estimated_delivery_time, created_at, order_status	Order information
Order_items	Order_item_id, order_id, item_id, quantity	Products and quantities in orders
Events	event _id, order_id, event_type, event_time_min, event_specific attributes	Generated occurrences

These are the logical items required for the application. They should be checked again with the actual SQLite Schema before treating this as a database specification.

Required Reference constraints:

- Each order must belong to exactly one student
- Each item must belong to exactly one seller
- Each order-item record must refer to an existing order and item
- Each seller-related event must refer to a seller participating in that order
- Each assigned driver must have a valid driver identity
- Every order_id must remain consistent throughout its event sequence
- All generated event identifier must be unique
- A successful order cannot contain unavailable items in its preparation workflow

The LLM must generate valid identifiers and relationships on its own. It must now rely on a supplied table of completed event examples.

Event type specifications:

Event type	Required event-specific attributes	Event description
OrderStarted	student _id, delivery_location, estimated_delivery_time, created_at	Begins an order
AddItem	Order_item_id, item_id, seller_id, quantity	Adds one item entry
OrderPlaced	Student_id, delivery_location, estimated_delivery_time, created_at	Submits the order
PaymentCompleted	Student_id	Records successful payment
PaymentFailed	student _id	Records failed payment
OrderCancelled	student _id	Terminates an unsuccessful order
ItemsPreparing	seller _id, is_available	Begins seller preparation
ItemsReady	seller_id	Completes seller preparation
DriverAssigned	driver_id	Assigns a driver
ItemsPickedUp	Seller_id, driver_id	Records pickup from one seller
OrderDelivered	student _id, driver_id, delivery_location, delivery_time	Completes delivery

Common fields every event carries: event_id, order_id, event_type, event_time_min

Workflow specifications:

1. Order started
2. Add item
3. Order placed
4. Payment completed
5. Seller workflow / Driver workflow
6. ItemsPickedUp
7. OrderDelivered

observation : All pickups must be completed before delivery

Constraints:

WF-01: Each complete order contains exactly one OrderStarted event

WF-02: Each order should contain at least one Item

WF-03:OrderPlaced occurs after the item's addition events

WF-04: Payment occurs after order placement

WF-05: An order cannot have both successful and payment outcomes

WF-06:Seller preparation begins only after successful payment

WF-07: Each participating seller must have corresponding preparation, readiness, and pickup events

WF-08:A driver must be assigned before pickup

WF-09:Sellers may prepare items concurrently

WF-10: An order cannot be delivered before every participation seller's item are collected

WF-11: A failed payment leads to cancellation, not delivery

WF-12:A completed order has exactly one terminal event: OrderDelivered or OrderCancelled

7. **Temporal specifications for event relationships:**

Relationship	Minimum Delay	Maximum Delay
OrderStarted->AddItem	1 MIN	5 MIN
OrderStarted->OrderPlaced	6 MIN	10 MIN
OrderPlaced-> PaymentCompleted	4 MIN	9 MIN
PaymentCompleted->ItemsPreparing	1 MIN	4 MIN
ItemsPreparing-> ItemsReady	Distribution-defined	Distribution-defined

PaymentCompleted->DriverAssigned	3 MIN	6 MIN
ItemsReady-> ItemsPickedUp	6 MIN	13 MIN
PaymentCompleted->OrderDelivered	Defined by distribution	Defined by distribution
PaymentFailed->OrderCancelled	1 MIN	3 MIN

Distributions specifications:

It determines how certain events should happen in the simulation.The generator should follow the according values in this specification. The LLM should generate the event values by itself, without receiving examples of events generated by the classical generator.

Order Arrival distribution

The number of orders should vary depending on the time of day. It runs for 720 minutes, representing a 12-hour period from 8 AM to 8 PM.

Preparation Time distribution

After a payment is completed, each seller can start preparing the items from their part of the order.

The preparation time is measured as the difference between the ItemsPreparing and ItemsReady events for the same order and seller.

When an order involves multiple sellers, each seller should have its own preparation process. This means that the preparation events can occur at different times, while still following the same distribution requirements.

Delivery Time distribution

The delivery time represents the amount of time between a successful payment and the completion of the order.It is calculated as the difference between the PaymentCompleted and OrderDelivered timestamps for the same order.

Additionally, the delivery time must also follow the workflow constraints. For example, an order cannot be delivered before all sellers have finished preparing their items and the driver has picked them up.

Output Requirements:

GEN-01: Generate valid event instances using only the defined event types

GEN-02: Generate identifiers and timestamps without receiving event examples

GEN-03: Preserve relationships across all events belonging to the same order

GEN-04: Follow all mandatory temporal, causal, and database constraints

GEN-05:Preserve the specified probability distributions across eligible observations

GEN-06:Do not assume unspecified variables are uniformly distributed or independent.

GEN-07: For a request to generate 40 event records, not 40 orders. An incomplete order is permitted at the end of a fixed-length stream.

GEN-08: For a distribution evaluation, generate complete orders so that the preparation and delivery durations can be measured

GEN-09: Output one JSON object per event, in chronological order, using defined attributes.

GEN-10: Do not provide explanations, invented event types, or copied reference sequences.